Yuvraj Pratap Singh

Artificial Intelligence Engineer • Medical Imaging • MLOps • Generative AI
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PROFESSIONAL SUMMARY

AI Engineer, **2+ years** building end-to-end ML/AI systems in **healthcare AI, medical imaging, and clinical NLP** for clients in Sweden and the US. **89% AUC clinical cancer detection**; multimodal RAG and multi-agent systems; **2 IEEE publications** and a **design patent**.

TECHNICAL SKILLS

Programming: Python, C++, SQL, JavaScript, TypeScript, Go, C#
ML / DL: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, SHAP, XGBoost, A/B Testing
Computer Vision: OpenCV, YOLOv5, YOLOv7, DICOM Processing, Medical Image Analysis
Healthcare AI: Clinical NLP, X-Ray/CT Analysis, HIPAA-Aware Development, FHIR, Alation
GenAI / LLMs: LangChain, RAG Pipelines, FAISS, Pinecone, OpenAI API, Ollama, Prompt Engineering
MLOps / Cloud: Docker, Kubernetes, MLflow, Airflow, CI/CD, FastAPI, gRPC, AWS (EC2/S3/SageMaker), GCP, Azure, n8n, Make
Data / Infra: Apache Spark, Apache Kafka, Redis, PostgreSQL, MongoDB, Pandas, NumPy

PROFESSIONAL EXPERIENCE

Artificial Intelligence Engineer | **WINNIIO AB** · Sweden — Remote — May 2025 – Present

- Architected multi-agent autonomous hospital simulation (Python, LangChain) with parallel task scheduling, reducing simulation run-time by 35–40% across 8+ concurrent agent threads.
- Built clinical NLP pipeline for doctor–patient dialogue interpretation (90%+ NER accuracy); deployed OCR module that cut document-to-record processing lag from hours to <5 minutes.

Research Intern | **Open Health Systems Lab (OHSL)** · USA — Remote — May 2025 – Jul 2025

- Developed XGBoost breast cancer detection pipeline on 10,000+ patient records achieving 89% AUC, 87% sensitivity; implemented SHAP explainability reducing clinician AI-review time by ~25%.
- Curated metadata for 50,000+ healthcare records via Alation and co-authored research paper for OHSL global open-health data initiative.

AI / ML Engineer | **Neurobridge Tech Pvt. Ltd** · Delhi — Dec 2024 – May 2025

- Built 3 end-to-end ML pipelines (ingestion → feature engineering → training → deployment via FastAPI), cutting training-to-production cycle time by 30% and boosting model accuracy by 12%.

Developer Intern | **VizExec Transformation Pvt. Ltd** · Gurgaon — May 2022 – Jul 2022

- Built WebRTC video conferencing module supporting 1,000+ concurrent participants (<200ms latency) for a large-scale political constituency engagement platform.

KEY PROJECTS

Agentic Multimodal Colon Cancer AI | Mar 2026 · github.com/Yuvraj235/Agentic_Multimodal_Colon_Cancer_AI

- Engineered 6-agent multimodal pipeline (ResNet50 + EfficientNet-B4 + BioBERT) fusing images, clinical text, and tabular data for colorectal cancer detection — 99.53% test accuracy, F1 0.9946, AUC-ROC 1.0 on HyperKvasir dataset.

Real-Time Metal Sheet Defect Detection | Mar 2023 – Jul 2023 · github.com/Yuvraj235

- Built YOLOv5/YOLOv7 CV pipeline on 12,000+ annotated images; achieved 96.8% mAP and <50ms inference via mosaic augmentation and anchor-box tuning — reduced false positives by 22% vs. baseline.

Lex-Indic-Legal — AI Legal Assistance Platform | Mar 2026 · github.com/Yuvraj235/Lex-Indic-legal

- Built RAG-powered Indian criminal law assistant (Groq Llama-3.3-70B + ChromaDB) covering 18 BNS sections and 51 IPC-to-BNS mappings; generates a branded 9-page legal analysis PDF in <60 seconds with voice-to-text input.

PUBLICATIONS

AI Framework for Enhanced Detection of Colonic Abnormalities | IEEE, 2025 · ieeexplore.ieee.org/document/11584361

Accuracy Enhancement of UAV using AI Models in Defense | IEEE ICCETM 2025, Sydney · ieeexplore.ieee.org/document/11558039

EDUCATION

MTech – Artificial Intelligence | Amity University, Noida — Aug 2024 – May 2026 · CGPA: 8.3/10

BTech – Computer Science & Engineering | Amity University, Noida — Jul 2019 – May 2023 · CGPA: 7.1/10

CERTIFICATIONS & ACHIEVEMENTS

- **Registered Design Patent No. 435775-001** — AI-Based Medical X-Ray Radiology Machine (Designs Act, 2000, Govt. of India)
- **PR Head, Amity Data Science Club** (2 years; grew membership by 40%)
- Top Performer — Ethical Hacking Online Training